

- 3 Catalase is an enzyme that catalyses the breakdown of hydrogen peroxide, a toxic waste product of metabolism.

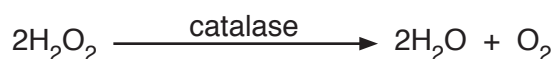


Fig. 3.1 shows the results of an investigation into the effect of hydrogen peroxide concentration on the rate of the catalase-controlled reaction, with and without the presence of two different inhibitors.

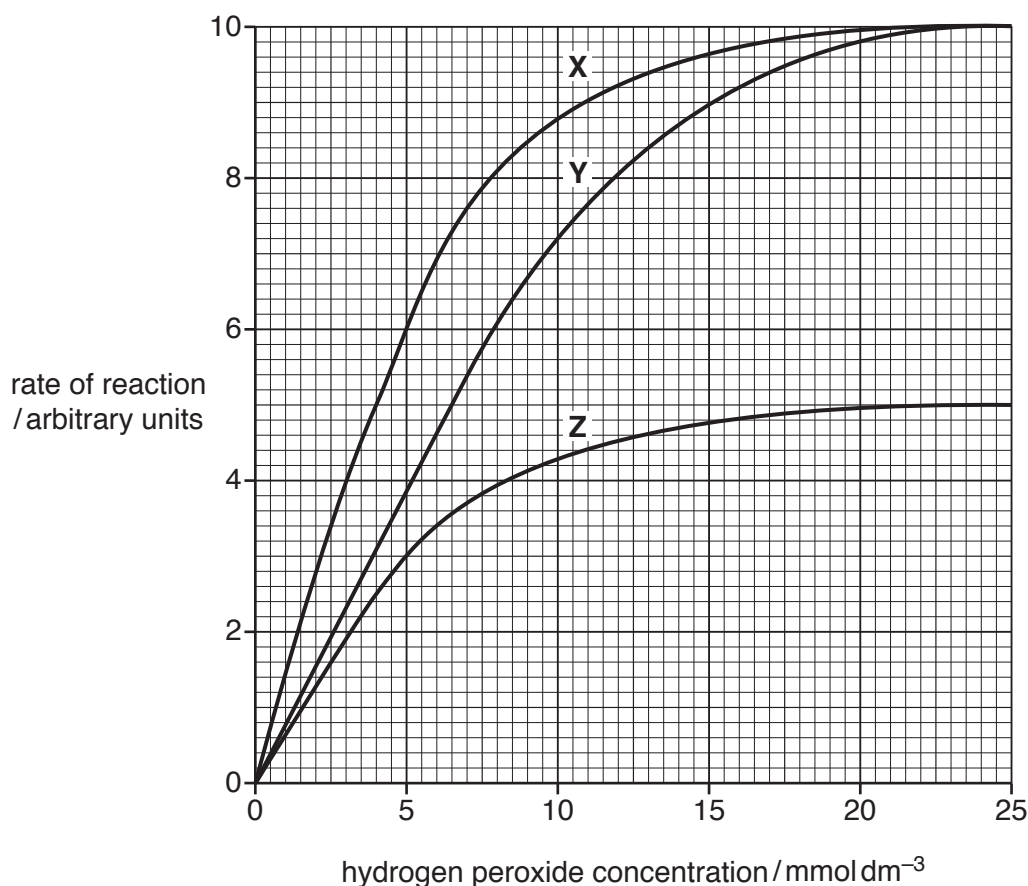


Fig. 3.1

- (a) The inhibitors used in the investigation have different modes of action.

Identify which of curves **X**, **Y** and **Z** are the results for:

- the reaction with the non-competitive inhibitor
- the reaction with the competitive inhibitor
- the reaction without any inhibitor.

non-competitive inhibitor

competitive inhibitor

without any inhibitor

- (b) With reference to Fig. 3.1, compare the maximum rate of reaction, $V_{\max}$, and the Michaelis-Menten constant, K_m , for curves **X**, **Y** and **Z**.

$V_{\max}$

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K_m

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[4]

Hydrogen peroxide has a harmful effect on cells. One effect is to damage DNA.

- (c) Describe the structure of DNA.

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- (d) The cell has mechanisms to repair the damage to DNA caused by hydrogen peroxide. Errors in repair may cause a change to the structure of DNA.

Studies have investigated possible risks associated with foods and drinks that contain hydrogen peroxide. This is because the compound can be considered a mutagen. Mutagens cause mutations.

Explain why hydrogen peroxide can be considered a mutagen.

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